

Escapade

ADAPTIVE & INCLUSIVE ALL-IN-ONE TRAVEL PLATFORM

Lifestyle Track: Planning an Escape (Travel Planner)

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1. Introduction

Escapade is an adaptive, all-in-one mobile travel application that unifies the end-to-end travel journey from AI itinerary creation and booking aggregation to multi-currency group budgeting and live mid-trip re-routing while incorporating Escapade AccessiRoute, a dedicated engine ensuring complete travel accessibility for individuals with physical, sensory, and cognitive disabilities.

While existing apps solve single fragments of a trip (e.g., booking or budgeting), Escapade provides a synchronized lifecycle engine that dynamically adapts when plans change and guarantees full inclusion for travellers with accessibility needs.

2. Problem Context & Impact Analysis

2.1 Problem Statement

Planning and executing a trip currently force travellers to navigate a fragmented ecosystem of 5+ disconnected applications: booking engines, flight trackers, split-expense spreadsheets, note apps, and messaging threads. This fragmentation creates four major failure points:

1. **Context Switching & Friction:** Travelers must manually copy data across platforms, creating room for human error, missing reservations, and planning fatigue.
2. **Group Coordination Overhead:** Aligning individual schedules, preferences, and budgets across a group creates friction, asynchronous delays, and financial disputes.
3. **Mid-Trip Rigidity:** When a flight is delayed or weather forces a closure, platforms offer zero automated assistance to dynamically re-plan remaining itinerary stops without disrupting existing bookings.
4. **Accessibility Invisibility:** Over 1.3 billion people globally live with disabilities. Mainstream travel apps omit critical accessibility metadata such as operational elevator status, ramp inclines, tactile guidance, or quiet sensory zones leaving disabled travellers at risk of unexpected physical or environmental barriers.

2.2 Target Audience & Reach

Target User Group	Specific Needs & Pain Points	How Escapade Solves It
Group Travelers & Friends	Budget misalignment, debt tracking, conflicting interest sets.	Shared preference matching, real-time group itinerary sync, automated multi-currency bill splitting.
Solo Adventurers	Safety concerns, complex logistics, overwhelming planning steps.	1-click AI itinerary generator, live travel alerts, integrated packing and visa checks.
Travelers with Physical Disabilities	Broken elevators, steep steps, narrow doorways, inaccessible transit.	AccessiRoute Engine: Verified step-free routing, door-width indicators, roll-in shower verification.
Neurodivergent Travelers	Sensory overload, unpredictable crowds, sudden schedule changes.	Sensory Radar predicting noise/crowd density; low-stimulation alternative suggestions.

2.3 Effectiveness: Before vs. After Escapade

- **Before:** Juggling Skyscanner, Excel, WhatsApp, Google Maps, and Splitwise. Unexpected flight delay causes missed activity bookings and lost money. Disabled travellers face physical barriers with no alternative routes.
- **After Escapade:** Single unified app. Flight delay triggers 1-tap AI itinerary re-adjustment. AccessiRoute continuously monitors transit accessibility in real time to reroute around physical barriers.

3. Ideation, Evolution

3.1 Exploration & Alternatives Considered

Before selecting Escapade's final architecture, three distinct product directions were generated and evaluated:

1. Idea A:

- **AI Travel Assistant Chatbot.** Simple prompt-based bot generating Markdown itineraries. Evaluated & Dropped: Too generic; lacks booking integration, expense splitting, and offline reliability.

2. Idea B:

- **Social Travel & Expense App.** Focused strictly on group decision-making and bill splitting. Evaluated & Dropped: Failed to solve live execution or mid-trip disruptions.

3. Idea C (Selected - Escapade):

- **Unified Travel Lifecycle Platform + AccessiRoute Engine.** An end-to-end mobile application covering Plan, Book, Manage, and During Trip, backed by an accessibility layer.

4. System Architecture & Feature Specifications

4.1 Pillar Breakdown

Pillar 1: Plan Trip

- **Trip Planner & AI Itinerary:** Generates day-by-day schedules optimized by budget, travel style, pace, and energy levels.
- **Trip Recommendations:** Context-aware activity suggestions based on weather and time of day.
- **Visa Check & Packing List:** Automated entry requirement checks based on traveller citizenship alongside smart, weather-adjusted packing checklists.

Pillar 2: Booking Hub

- **Unified Aggregator:** Aggregates search and booking flows across Flights, Stays (Hotels/Hostels), Ground Transport (Trains/Buses/Rideshare), and Local Activities into a single checkout flow.

Pillar 3: Manage Trip

- **Budget Tracker & Multi-Currency Ledger:** Real-time expense monitoring converted instantly into the group's primary home currency.
- **Bill Splitter & OCR Receipt Scanner:** Snap a photo of any receipt; computer vision extracts itemized line items and assigns costs to specific group members.

Pillar 4: During Trip

- **Live Flight & Travel Alerts:** Real-time push notifications regarding gate changes, flight delays, weather warnings, and local safety advisories.
- **Dynamic AI Re-Planner:** 1-tap schedule adjustment if an event is delayed or cancelled.

4.2 Speciality Innovation: Escapade AccessiRoute

- **Step-Free Barrier-Aware Routing:** Live map navigation filtering out stairs, steep inclines (>8%), unpaved paths, and out-of-service elevators using civic telemetry and user reports.
- **Sensory Environment Radar:** Predicts venue decibel levels, crowd density, and lighting conditions to recommend sensory-friendly visiting windows for autistic or neurodivergent travellers.
- **AI Sign-Language & Menu Assist:** Camera-based AI that interprets basic sign language gestures into text/speech and reformats dense, stylized menus into dyslexic-friendly text.
- **Caregiver Expense Ledger:** Dedicated accounting module for tracking assistant tickets, medical equipment baggage allowances, and subsidized fares separately from standard group expenses.
- **1-Tap Medical SOS Sync:** Instantly broadcasts precise GPS coordinates, blood type, emergency contacts, and medical conditions to local emergency services and designated travel companions.



Figure 4.2.1: AccessiRoute compares a normal route against a step-free, elevator-verified alternative in real time.

5. Technical Feasibility & Tech Stack

5.2 Implementation Feasibility & Resource Realism

- **Cross-Platform Delivery:** Flutter enables a single codebase deployment across Android and iOS while enforcing native screen-reader support (TalkBack/VoiceOver).
- **API Cost Optimization:** Caching static place details and itinerary structures in Firebase drastically reduces live API call volume to external providers.
- **Offline Readiness:** Critical itinerary data, reservation tickets, and offline barrier maps are stored locally on-device using SQLite/Hive, ensuring functionality even in low-connectivity areas.

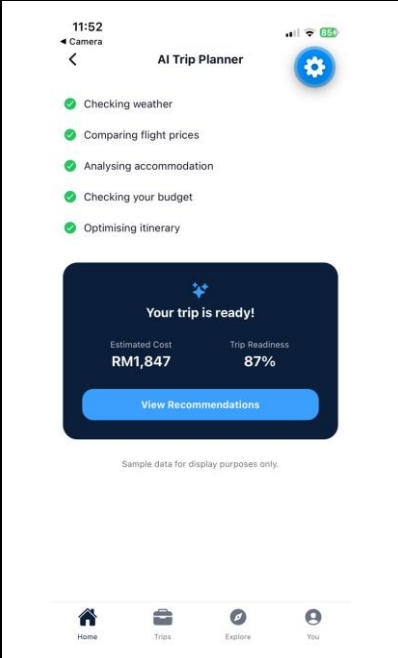
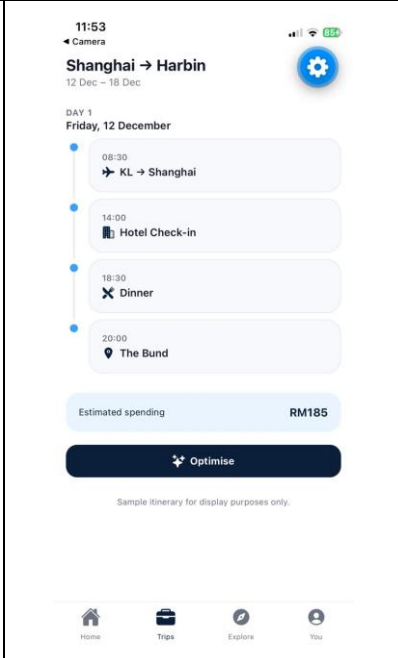
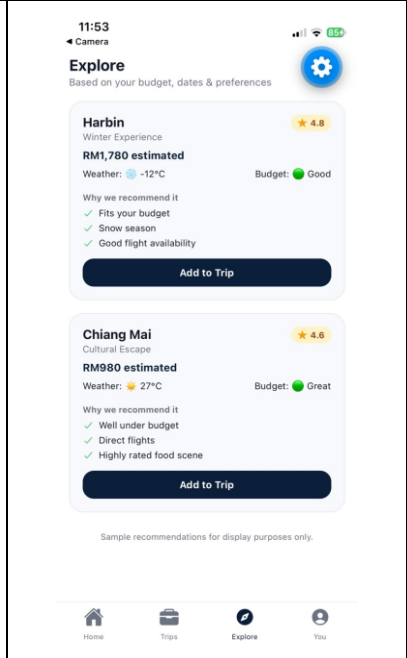
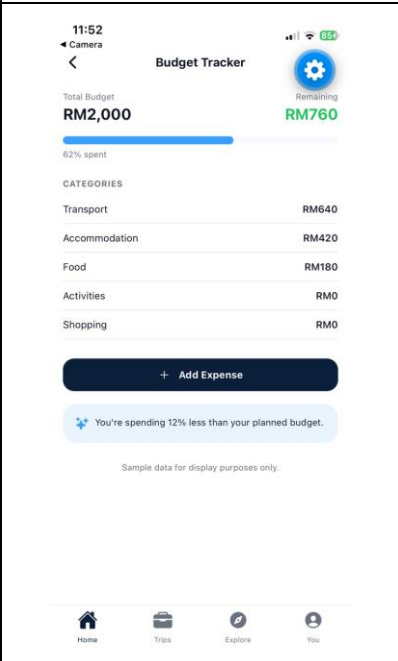
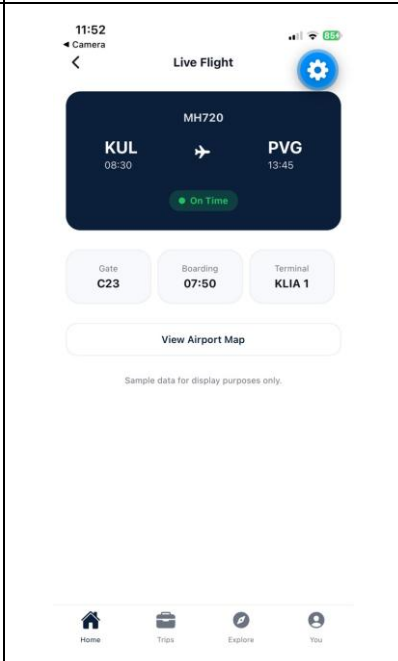
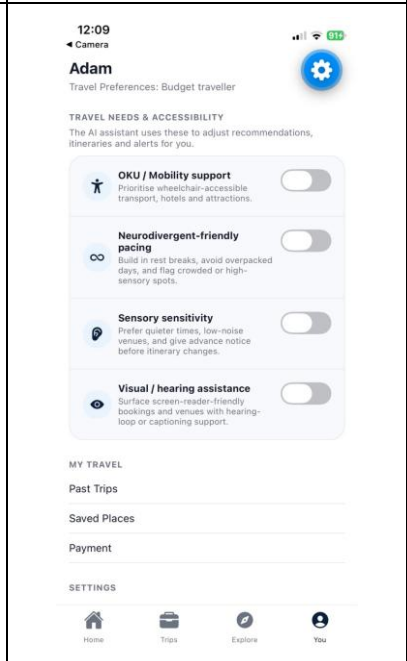
6. UX/UI Design & Accessibility Framework

6.1 Design Principles

- **WCAG 2.1 AA Compliance:** High-contrast colour ratios (minimum 4.5:1 for standard text, 7:1 for high-contrast mode).
- **Touch Target Sizing:** Minimum touch targets set to 48x48 dp across all interactive elements.
- **Readable Typography:** Dynamic font sizing supporting OpenDyslexic and system-level font scaling.

6.2 Prototype Screens

The following screens from the working prototype demonstrate the design principles above applied to the core Plan, Book, Manage, During Trip, and personalization flows.

 <i>AI Trip Planner: readiness scoring, dynamic feedback</i>	 <i>Day-by-day itinerary: clear timeline, high contrast</i>	 <i>Explore: card-based, reasoned recommendations</i>
 <i>Budget Tracker: 48x48dp targets, 4.5:1 contrast</i>	 <i>Live Flight: high-contrast status alerts</i>	 <i>Accessibility profile: OKU, sensory & neurodivergent toggles</i>

7. Diagram

7.1 Architecture Diagram

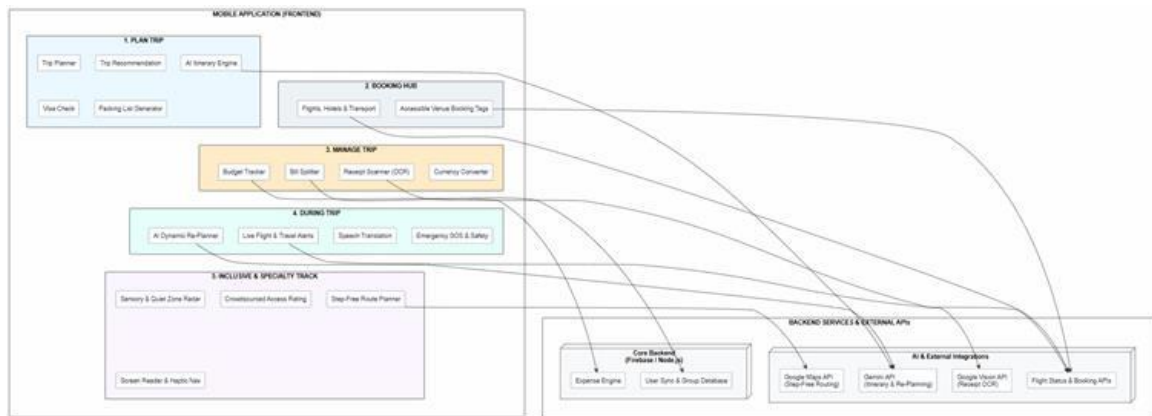


Figure 7.1.1: System architecture - mobile frontend pillars, backend services, and external API integrations.

7.2 Sequence Diagram

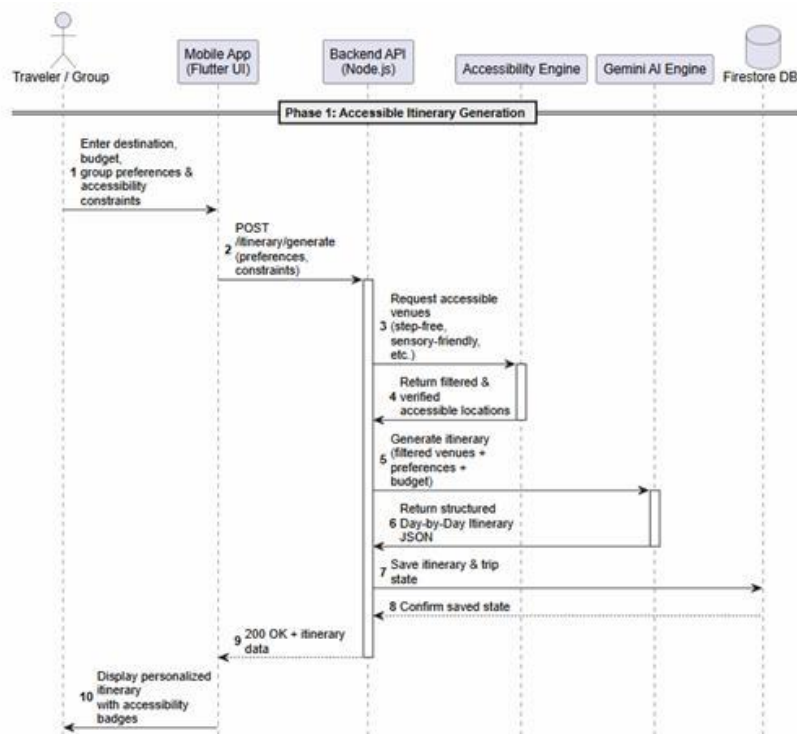


Figure 7.2.1: Phase 1 - accessible itinerary generation.

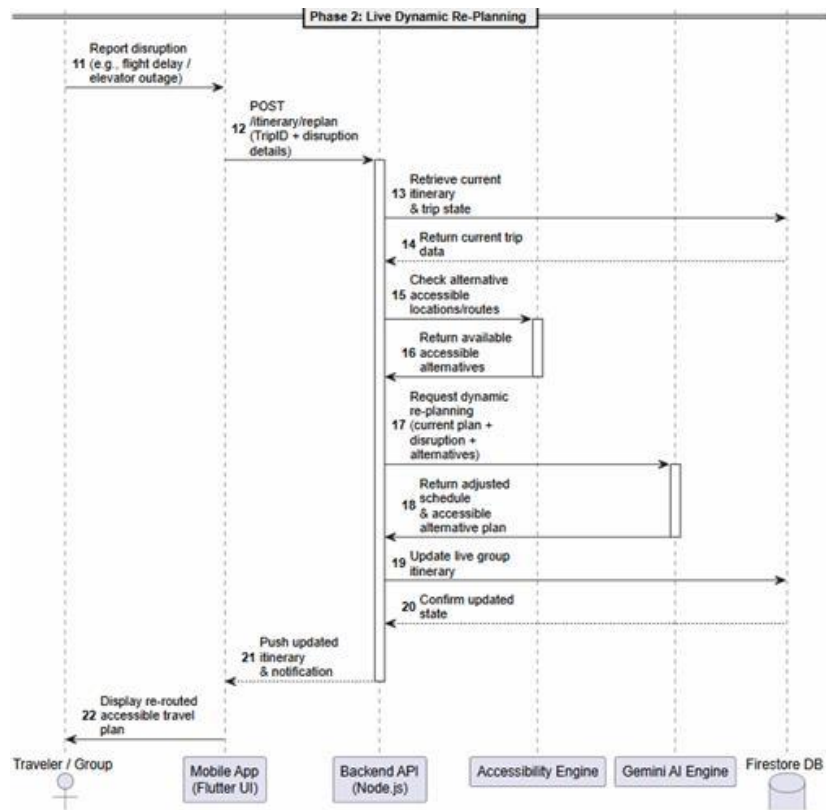


Figure 7.2.2: Phase 2 - live dynamic re-planning.

7.3 Class Diagram

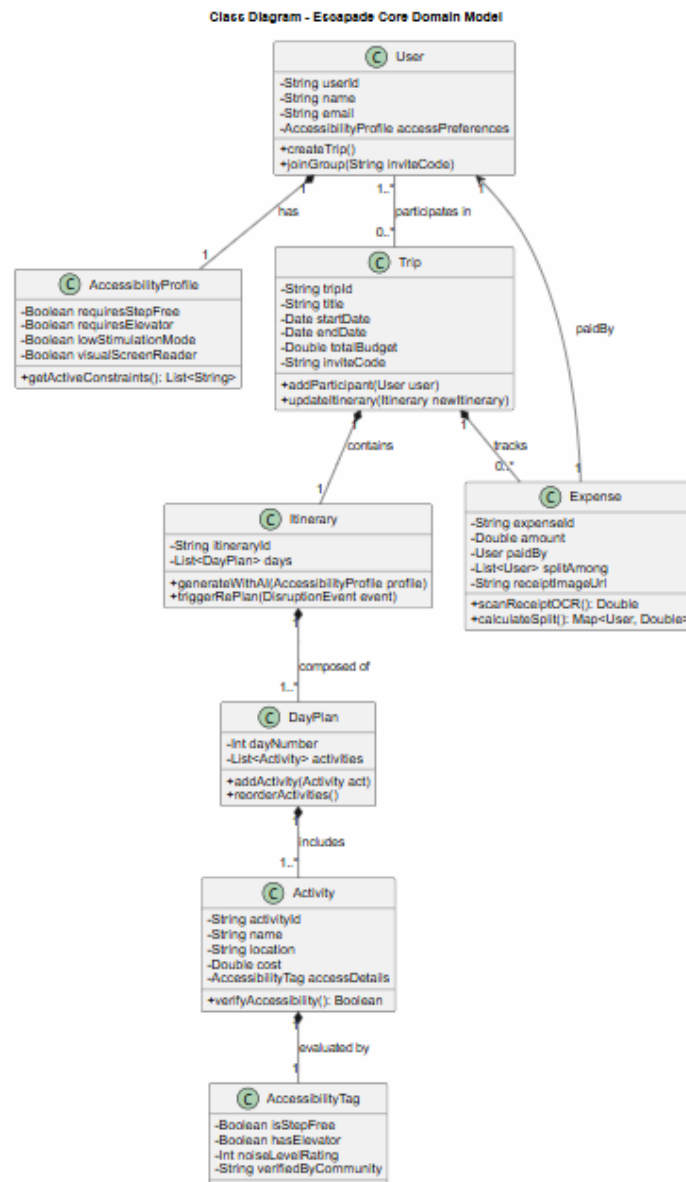


Figure 7.3.1: Escapade core domain model.